THE FLOWING MATERIAL BALANCE PROCEDURE

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The "Flowing" Material Balance Procedure

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ABSTRACT

The pressures used in constructing the traditional material balance (p/z) plot must be fully built-up reservoir pressures, usually obtained by shutting-in the well. The procedure described in this presentation does NOT require the well to be shut-in. Instead, it utilizes information obtained from standard good operating practices (production and flowing pressure) to quantify the gas-in-place, without having to shut-in the well.

The classical pseudo-steady state analysis and its shortcomings are discussed. In addition, a new procedure known as the "flowing" Material Balance is introduced. This procedure consists of a p/z plot of the "flowing" pressure (as opposed to the "shut-in average" reservoir pressure) versus cumulative production. A straight line can be drawn through the data and then, a parallel line, drawn through the initial reservoir pressure, gives the initial gas-in-place. Variations of this method, using wellhead pressures (tubing and casing) are discussed, along with field examples. The method is a very PRACTICAL and powerful tool for the early quantification of reserves.

INTRODUCTION

The determination of gas reserves is a fundamental calculation in reservoir engineering. This information is of critical importance in determining production strategy, design of facilities, contracts and the value of the reserves. Reserves can be estimated in three ways — Volumetric, Production Decline and Material Balance. The Production Decline gives an estimate of recoverable gas, whereas the other two give an estimate of gas-in-place. In the reservoir

engineering literature, the word "reserve" refers to the raw gas-in-place, whereas in the commercial world, the word "reserve" often means the recoverable sales gas. In this paper, the word "reserve" is used to denote the raw gas-in-place.

Volumetrically determined reserves can be very imprecise, because they depend on the areal extent of the pool, which is often unknown. Production Decline methods give the "recoverable" gas volumes under the existing operating conditions. A change in these operations, for example a lowering of the compressor suction pressure, can change the magnitude of the recoverable gas. The original gas-in-place is difficult to ascertain. The Material Balance method (given well-behaved data) gives the most accurate estimate of original gas-in-place. Once this value is known, it can be used to forecast the recoverable gas under various operating scenarios.

CLASSICAL MATERIAL BALANCE

The classical material balance expresses a relationship between the average pressure in the reservoir and the amount of gas produced. When there has been no production, the pressure equals the initial reservoir pressure, when all the gas has been produced, the pressure in the reservoir is zero. This assumes that the reservoir acts like a tank and there is no external pressure maintenance. The relationship between pressure and cumulative production is approximately linear. If the compressibility factor z, is taken into account, then the material balance plot of p/z versus cumulative production, Q, is a straight line going from the initial pressure p_i/z to the initial gas-in-place, IGIP. Deviations from this straight line can be caused by external recharge or offset drainage. For the purposes of this presentation, we are only considering reservoirs which have straight line material balance plot of p/z versus Q. The more comprehensive material balance procedures for complex reservoirs has been treated extensively by Havlena and Odeh. [1]

In order to generate a p/z plot, the well is shut-in at several points along its producing life, and its average reservoir pressure is obtained from a properly conducted buildup test and interpretation. Often, the duration of the shut-in is not long enough, and the build-up data have to be extrapolated and corrected to obtain the average reservoir pressure. This is one of the causes of erratic data often observed in the p/z versus Q plots. A straight line through the initial pressure point, and going through the data can be extrapolated to p/z equal to zero to give the initial gas-in-place on the Q axis.

SHUT-IN/FLOWING DATA

The classical material balance procedure described above depends upon shut-in static pressures. It is possible to obtain information on reservoirs from flowing pressure data, under certain circumstances. These procedures form the crux of this paper.

The flow of gases through porous media can be divided into two major categories — Transient and Stabilized. Transient flow behavior is dominated by reservoir characteristics such as permeability, skin, heterogeneities, location of boundaries, etc... and a complex function of time (log time). Stabilized flow, on the other hand, is dominated by reserves and a simple time function (t).

High and medium permeability reservoirs reach stabilized flow relatively quickly — within a few weeks — but low permeability reservoirs can take a year or more to stabilize. Because stabilized flow is dominated by reserves, it should be possible to estimate the reserves in a pool, if flow rate pressure and time data were available during the period of stabilized flow. Thus for medium and high permeability

reservoirs, we can calculate the reserves from flowing data without having to shut-in the well.

The remainder of the paper reviews established methods of doing this and presents more practical alternatives. The discussions that follow assume a single well pool, but the principles can easily be extended to multi-well pools.

When a well is in stabilized flow, its behavior is represented by the pseudo-steady state equations found in the well testing literature.[2] At the heart of all these equations lies the material balance equation, in one form or another. The following discussion will describe several procedures for determining the initial gas-in-place from flowing data. All these methods are variations of the material balance equation, with different assumptions and approximations.

PRODUCTION DATA

The various methods of analysis will be illustrated by data from a producing well. For the first two months, the well was flowing at a fairly constant rate of approximately 250 $10^3 m^3/d$. The tubing and casing pressures were measured daily. The casing pressures were also converted to bottomhole pressures, using a multi-step Cullender and Smith [3] calculation.

A review of the well test data and the core analysis indicates that this formation possesses a good permeability to gas (50 md). The time to stabilization (assuming 1 section spacing) is two weeks approximately. This means that the bulk of the production data represents stabilized (pseudo-steady) flow.

Method 1: Classical Pseudo-Steady State Analysis (PSS)

This is the standard method of pseudo-steady analysis described in the well testing literature [2][4]. A plot is made of pressure squared (or pseudo pressure) versus time on cartesian coordinates. A straight line is drawn through the appropriate portion of the data corresponding to constant rate production, and the slope of that straight line is used to calculate the gas-filled pore volume, and hence, the initial gas-in-place (IGIP) (ERCB Guide G-3, variation of Equation 4-29 in SI units)[4].

$$\text{IGIP} = \frac{2.929 \times 10^{-2}\ q \times 10^{3}\ zTp_i\ 288}{c\ \text{slope}/24\ z_i\ T\ 101.325}$$

$$= \frac{2 \times 10^{-3}\ q\ p_i}{c\ \text{slope}}\quad 10^6 m^3$$

Figure 1 shows a plot of pressure squared (casing pressure converted to sandface) versus time in days. The slope of this line is 780 000 kPa^2/day. Substituting in the above equation gives:

$$\text{IGIP} = \frac{2 \times 10^{-3} \times 250 \times 12900}{1.3 \times 10^{-4} \times 780000}$$

$$= 64\ 10^6 m^3$$

This is the classical pseudo-steady state analysis procedure. It assumes that the compressibility, c, is constant. This assumption is good for an undersaturated liquid but not so good in gas. In a gas reservoir c varies as 1/p, but in order to apply this pseudo-steady state procedure, c is traditionally assumed to be constant at initial condition. This can be a significant source of error as depletion progresses.

In the above calculations c was calculated at the average pressure of 8500 kPa measured during the 2 months of production. If the initial pressure had been used for calculating c (12900 kPa), the IGIP would be 94 $10^6 m^3$. It is obvious that this calculation is highly dependent on the gas compressibility (which varies significantly with pressure). An additional complication arises when the rate is not constant. In such a case, the pressures should be rate normalized, and possibly converted to pseudopressures as well.

Method 2: "Flowing" Material Balance (FMB)

A clear understanding of what pseudo-steady state is can provide a new insight into calculating reserves by a procedure similar to the traditional material balance (p/z vs Q) plot. The difference is that this method utilizes the flowing pressure rather than the shut-in pressure. When a reservoir is in pseudo-steady state flow, the pressure at all locations in the reservoir declines at the same rate. This is illustrated in Figure 2, which depicts pressures in the reservoir at all locations - the wellbore on the left, and the exterior boundary of the reservoir on the right. Each of the lines 1, 2 and 3 represents the pseudo-steady state pressure in the reservoir when the well is flowing at a constant rate. The sketch illustrates that the pressure decline from time 1 to time 2, is the same throughout the reservoir (the curves are "parallel"). The same is true between curves 2 and 3.

The traditional Material Balance (p/z) procedure would be to shut-in the well at time 1, and let the pressure in the reservoir stabilize. This would give an average reservoir pressure p_{R1}. Similarly, the pressure profile in the reservoir, at times 2 and 3 would result in shut-in average reservoir pressures of p_{R2} and p_{R3}. It is clearly evident that the drop in pressure from p_{R1} to p_{R2} to p_{R3} is the same as the drop in pressure at any selected point along the curves 1, 2 and 3 (the curves are "parallel"). A convenient point for pressure analysis is the pressure at the wellbore. Remembering that curves 1, 2 and 3 represent a flowing well, it becomes evident that a plot of the flowing wellbore pressure (Pwf1, Pwf2, Pwf3) should be parallel to a plot of the average reservoir pressure (p_{R1}, p_{R2}, p_{R3}).

The above discussion illustrates that a Material Balance (p/z vs Q) plot using the flowing wellbore pressures (Pwf), should be parallel to the Material Balance plot using the average reservoir pressure (p_R). This procedure, referred to as the "Flowing Material Balance" consists of a plot of p/z, where p is the flowing pressure at the wellbore (or for that matter any location in the reservoir) versus cumulative production. A straight line is drawn through the flowing pressure data and a parallel line is drawn through the initial reservoir to give the initial gas-in-place (IGIP). The "Flowing Material Balance" should be performed on the flowing sandface pressures. This has been done and is shown on Figure 3 resulting in an IGIP = 71 $10^6 m^3$. The sandface flowing pressures were calculated from a measurement of the casing pressure. Since the well was flowing through tubing the calculation from sandface (casing) pressure to bottomhole pressure was done using a static column of gas. It is clearly evident from Figures 1 and 3, that a plot for casing wellhead pressures has the same trend as the bottom hole pressures Thus a "Flowing Material Balance" plot was constructed using the casing pressures (instead of bottomhole pressures) see Figure 4. A line drawn parallel to the data through the initial "wellhead" pressure (NOT the initial "reservoir" pressure) gives an IGIP = 72 $10^6 m^3$.

This method depends on the reservoir being in pseudo-steady state flow. This means that the early time data should be ignored because transient flow is still dominant at that time.

Method 3: Tank Model

In gas deliverability forecasting, a "tank" model is often used. This type of program is based on the Material Balance equation in the reservoir (tank) and incorporates the deliverability potential of the well. It can be used for forecasting the rate at a given flowing pressure, or conversely, the pressure at a given rate. Such a program was used to model the flowing wellhead pressure for various magnitudes of gas-in-place. The well's flow rate was set to

a constant 250 $10^3 m^3/d$ and the decline in wellhead flowing pressure plotted for various IGIP. The results are shown in Figure 5. The best match to the measured data was obtained using a "tank" volume of 76 $10^6 m^3$. The range of IGIP = 70 to 90 $10^6 m^3$ is shown on the Figure. The slope of the lines drawn on Figure 5 is affected mostly by the IGIP and the flow rate.

Method 4: Approximate Wellhead Material Balance (AWMB)

If one recognizes that, often, the change in the compressibility factor, z, is small (it can be of the same order of magnitude as the errors in the data!), then ignoring z and plotting p vs Q rather than p/z vs Q can give a very reasonable approximation of the IGIP. The initial and end points $p = p_i$ and $p=0$ are correct, but theoretically, the line joining them is not necessarily straight. Dividing p by z makes it into a straight line. Given this understanding, and recognizing that the wellhead to bottomhole pressure difference is approximately constant (see Figure 1) a procedure similar to Method 2, (but ignoring z) is to plot the "wellhead" (casing) pressure versus cumulative production instead of bottomhole pressure. If the variations in the z-factor are small, a straight line will be obtained. A parallel line drawn through the initial static "wellhead" pressure (kPa) will give an estimate of the initial gas-in-place. For the data of this example, the plot is shown on Figure 6 and gives an IGIP = 71 $10^6 m^3$.

This is a simplified Material Balance plot based on wellhead pressures (rather than reservoir pressures) and it ignores z factor. Our experience with this type of analysis indicates that it is a very practical and acceptable procedure.

Method 5: Production Decline

A traditional production decline plot is shown in Figure 7. For the first two months, the rate was restricted. After that, a decline trend was obvious. Extrapolation of this trend gives recoverable reserves not IGIP. The recoverable reserves are 37 $10^6 m^3$ which is consistent with the IGIP calculated by the other method and reflects a recovery factor of approximately 50% under present operating conditions.

SUMMARY

Several methods have been presented for estimating the initial gas in place without shutting in the well. The calculated values are summarized below:

	IGIP ($10^6 m^3$)
Method 1 (PSS)	64
Method 2 (FMB) Reservoir	71
(FMB) Wellhead	72
Method 3 (Tank)	76
Method 4 (AWMB)	71
Average	71

CONCLUSIONS:

1. The initial gas-in-place determined from the flowing pressure/production data is approximately 70 $10^6 m^3$. This is significantly different from the volumetric calculations for a 1 section spacing (500 $10^6 m^3$), indicating a much smaller areal extent to the pool in which this well is completed.

2. The procedures presented in this paper provide a very PRACTICAL tool for estimating gas-in-place, using data generally available during production operations.

3. Material Balance Calculations using shut-in pressures were not available for this pool. Yet it was possible to determine the reserves with reasonable certainty, and have them confirmed by production decline analysis.

NOMENCLATURE:

c	=	gas compressibility, 1/kPa
IGIP	=	Initial Gas-In-Place $10^6 m^3$
p_i	=	initial reservoir pressure, kPa
Q	=	cumulative production $10^6 m^3$
q	=	gas rate, $10^3 m^3/d$
slope	=	kPa^2/day
T	=	temperature, °K
z	=	compressibility factor

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3. Cullender, M.H. and R.V. Smith: "Practical Solution of Gas Flow Equations for Well and Pipelines with Large Temperature Gradients", Trans. AIME, Vol. 207, P. 281

4. E.R.C.B "Gas Well Testing - Theory and Practice", Energy Resources Conservation Board, Alberta, Canada. 1979, Fourth Edition.

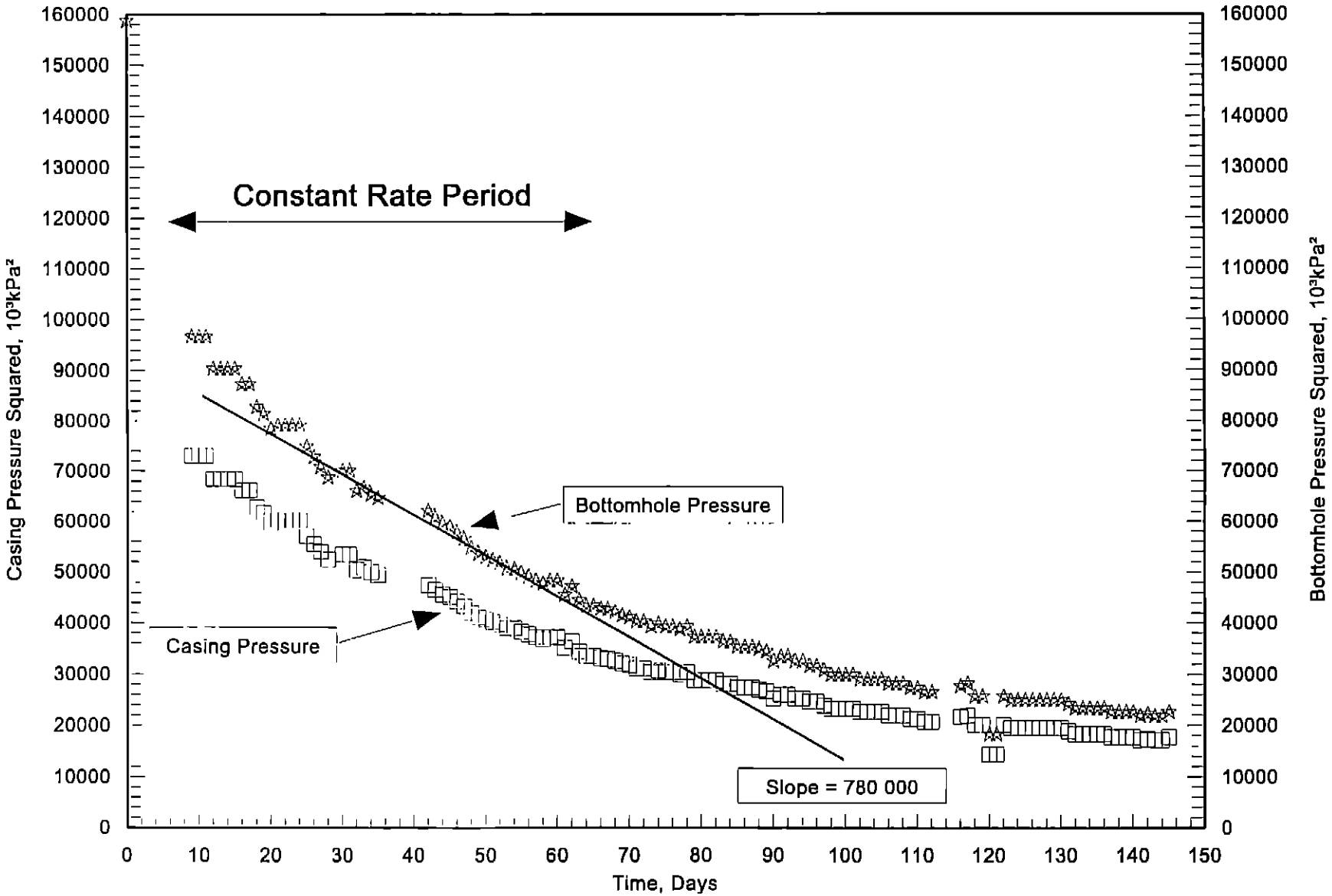

FIGURE 1: Pseudo-Steady State Analysis

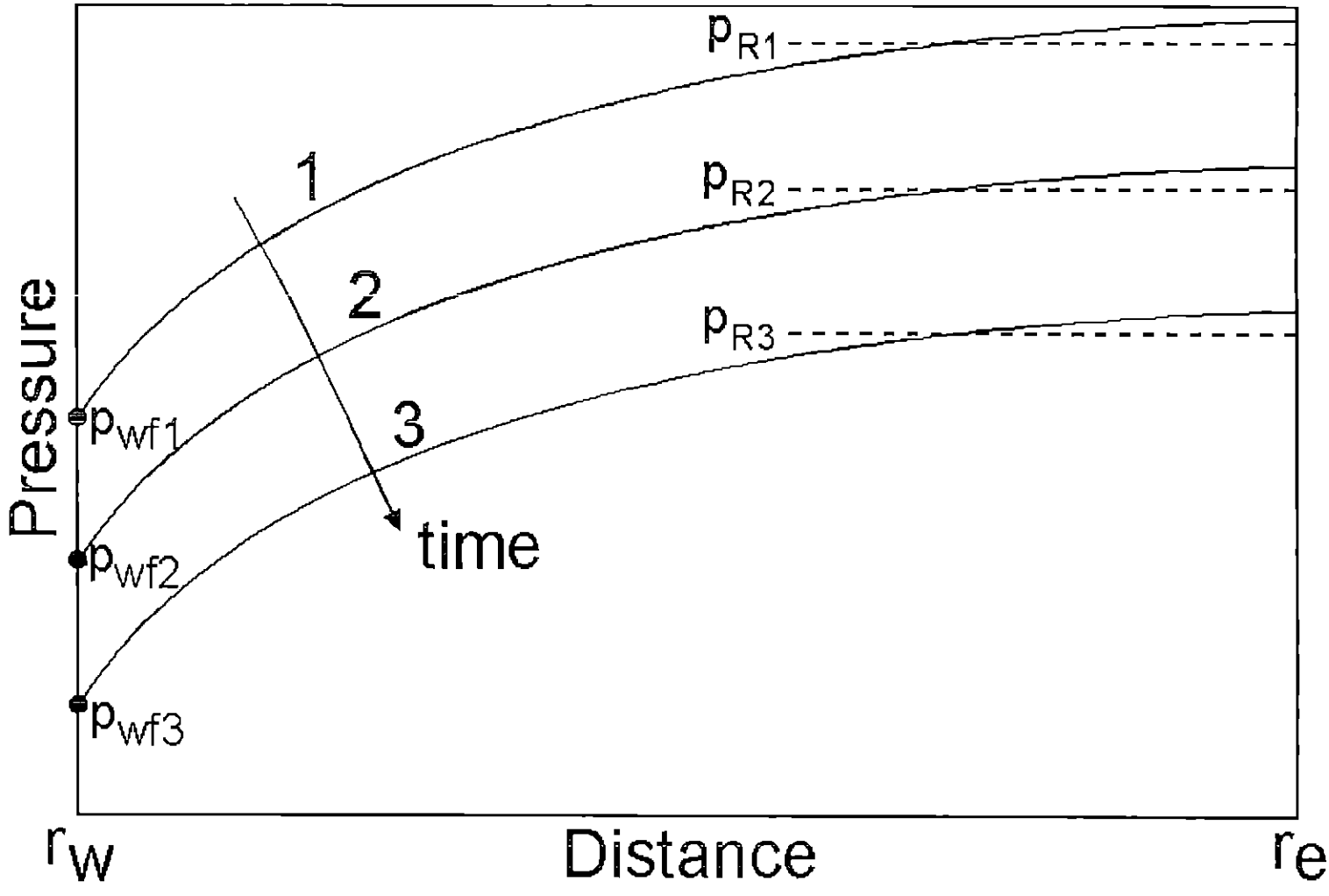

FIGURE 2: ILLUSTRATION OF PSEUDO-STEADY STATE

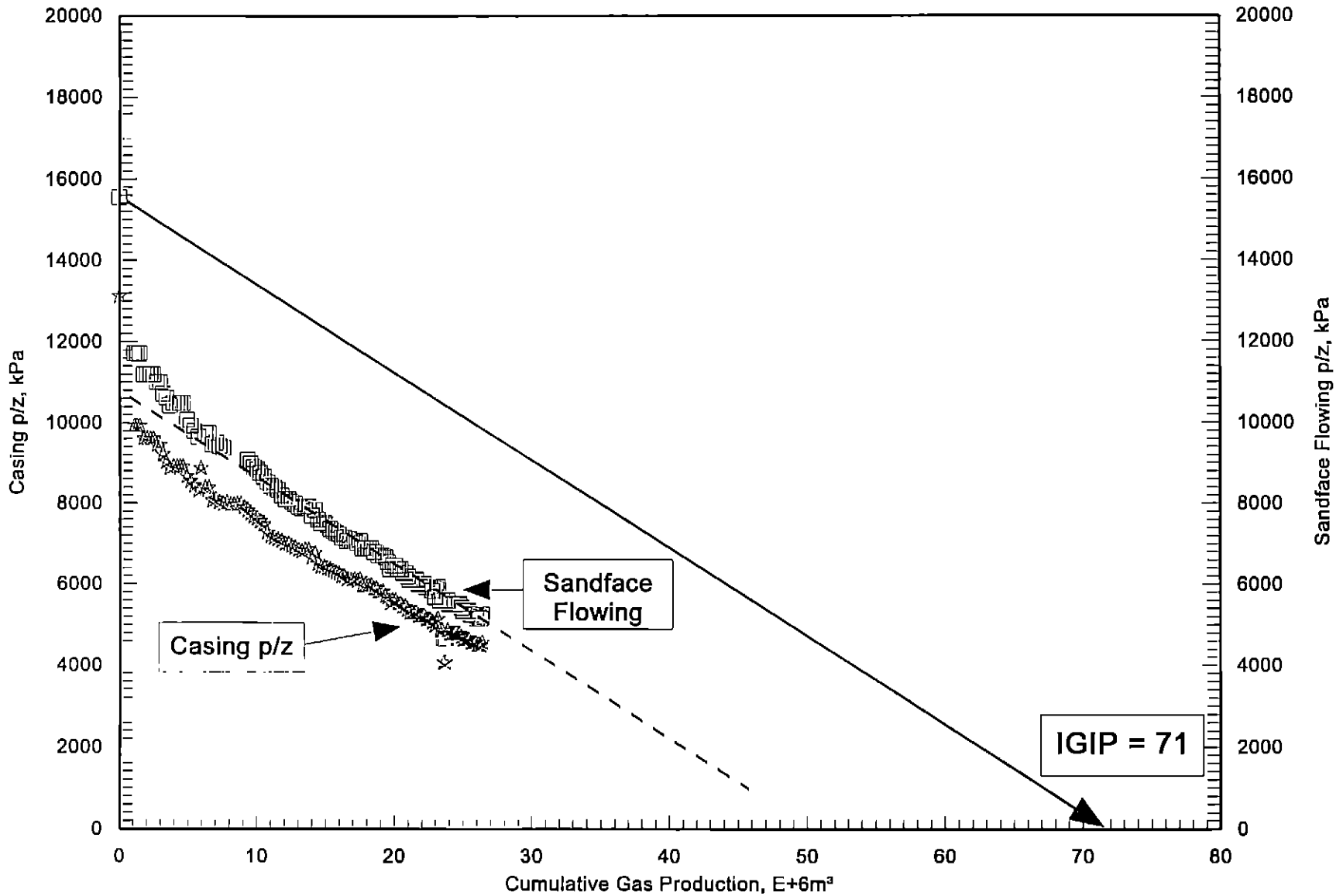

FIGURE 3: "Flowing" Material Balance

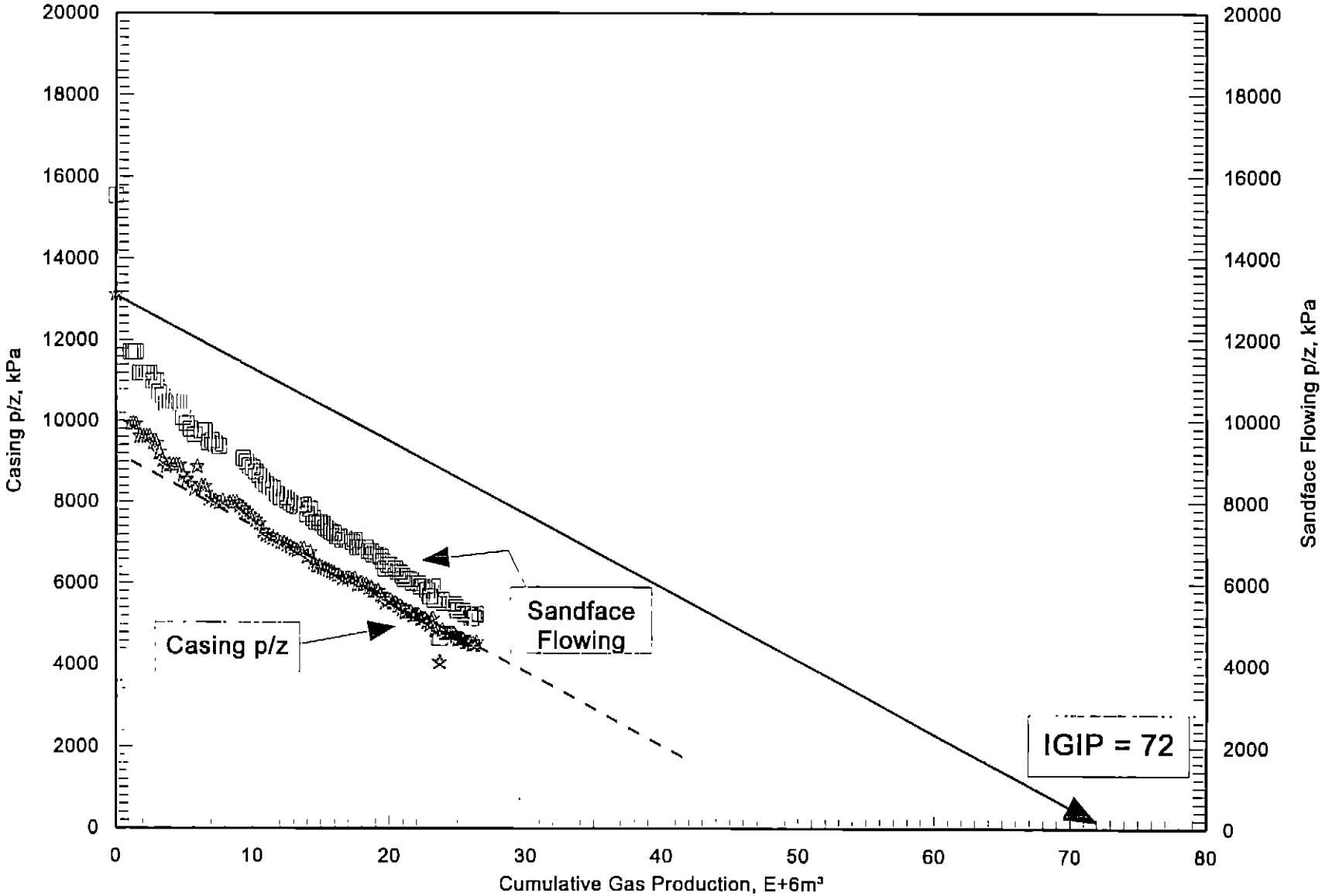

FIGURE 4: "Flowing" Material Balance (Casing Pressures)

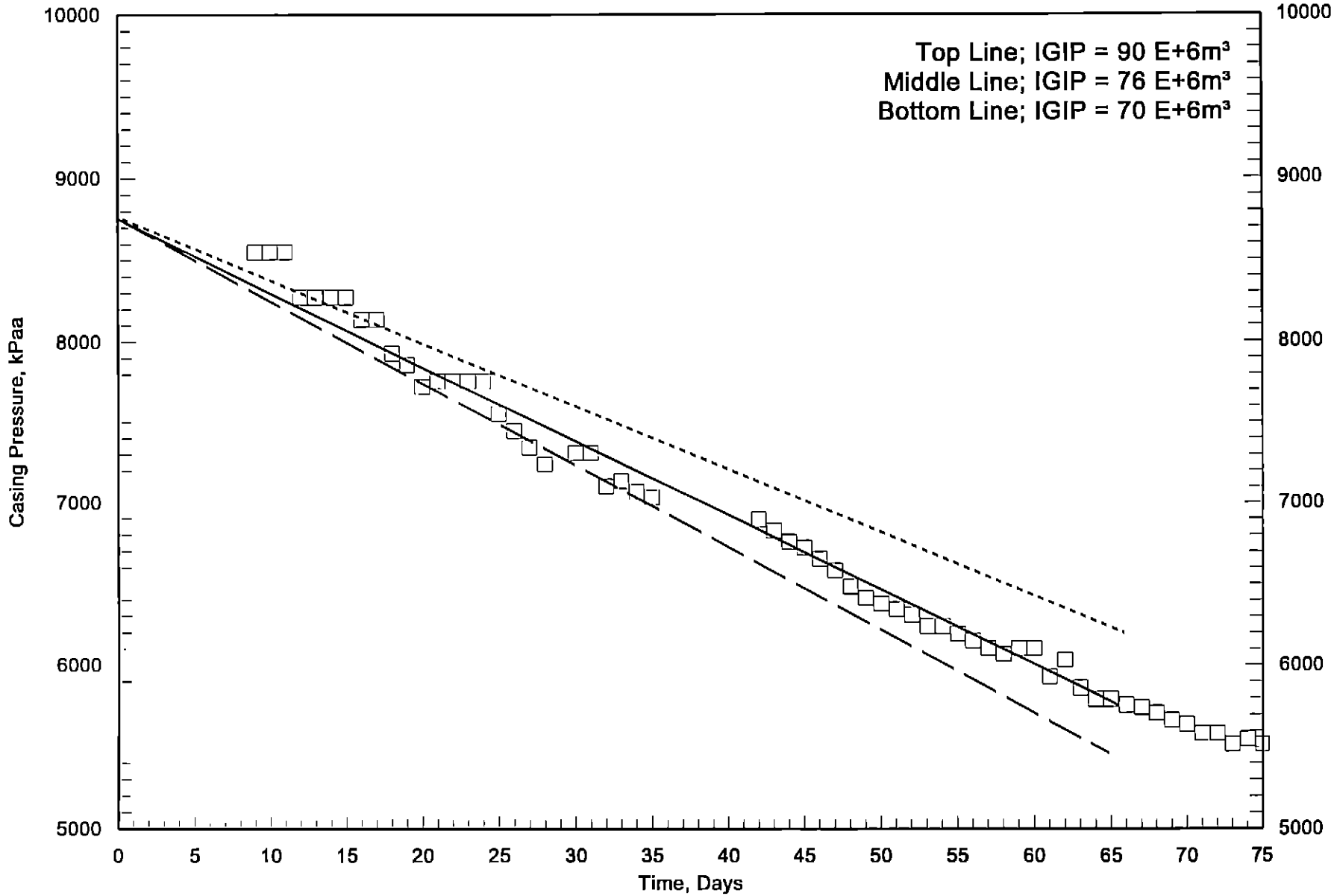

FIGURE 5: Simulated Flowing Pressure Decline using Tank Model

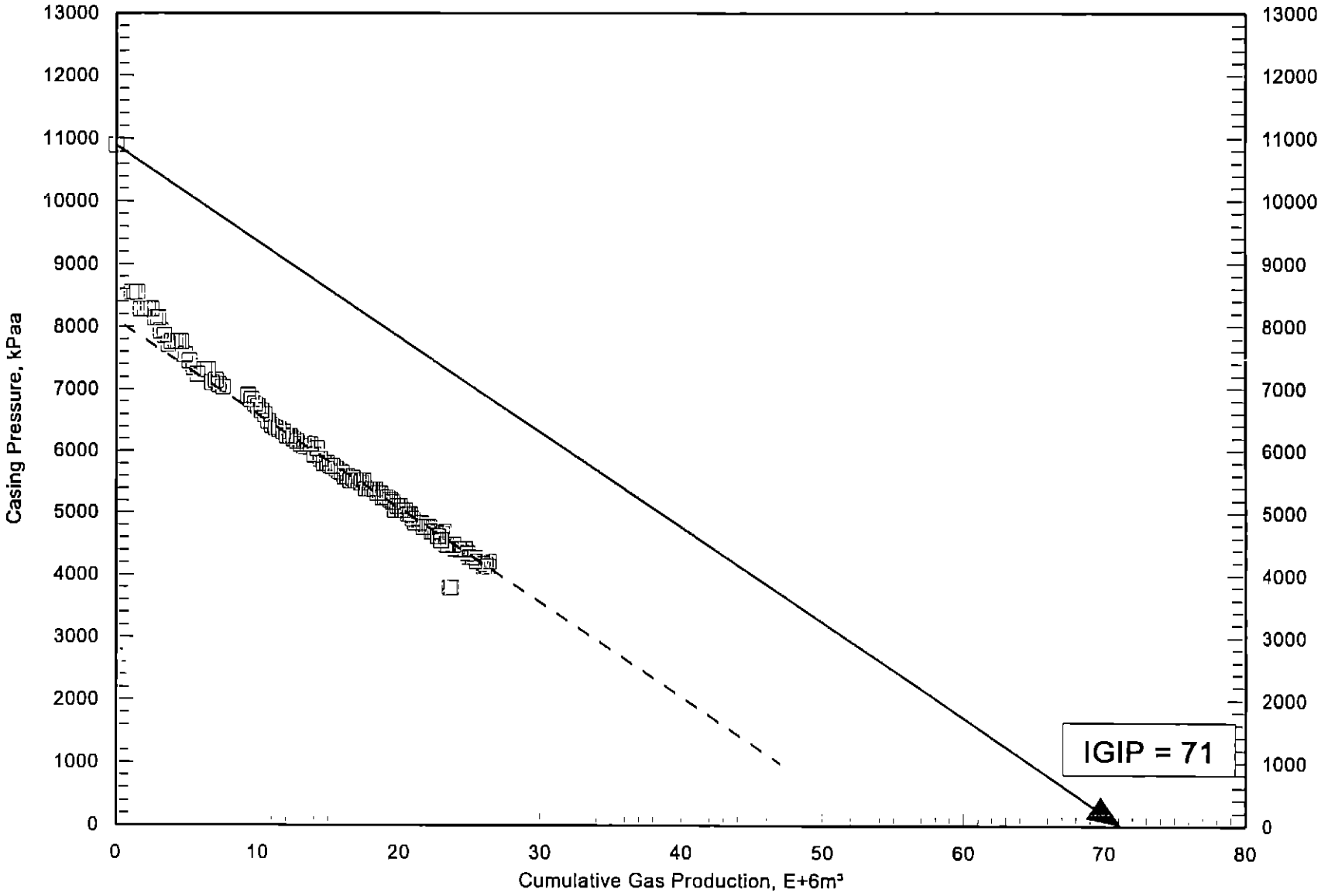

FIGURE 6: Approximate Wellhead "Flowing" Material Balance

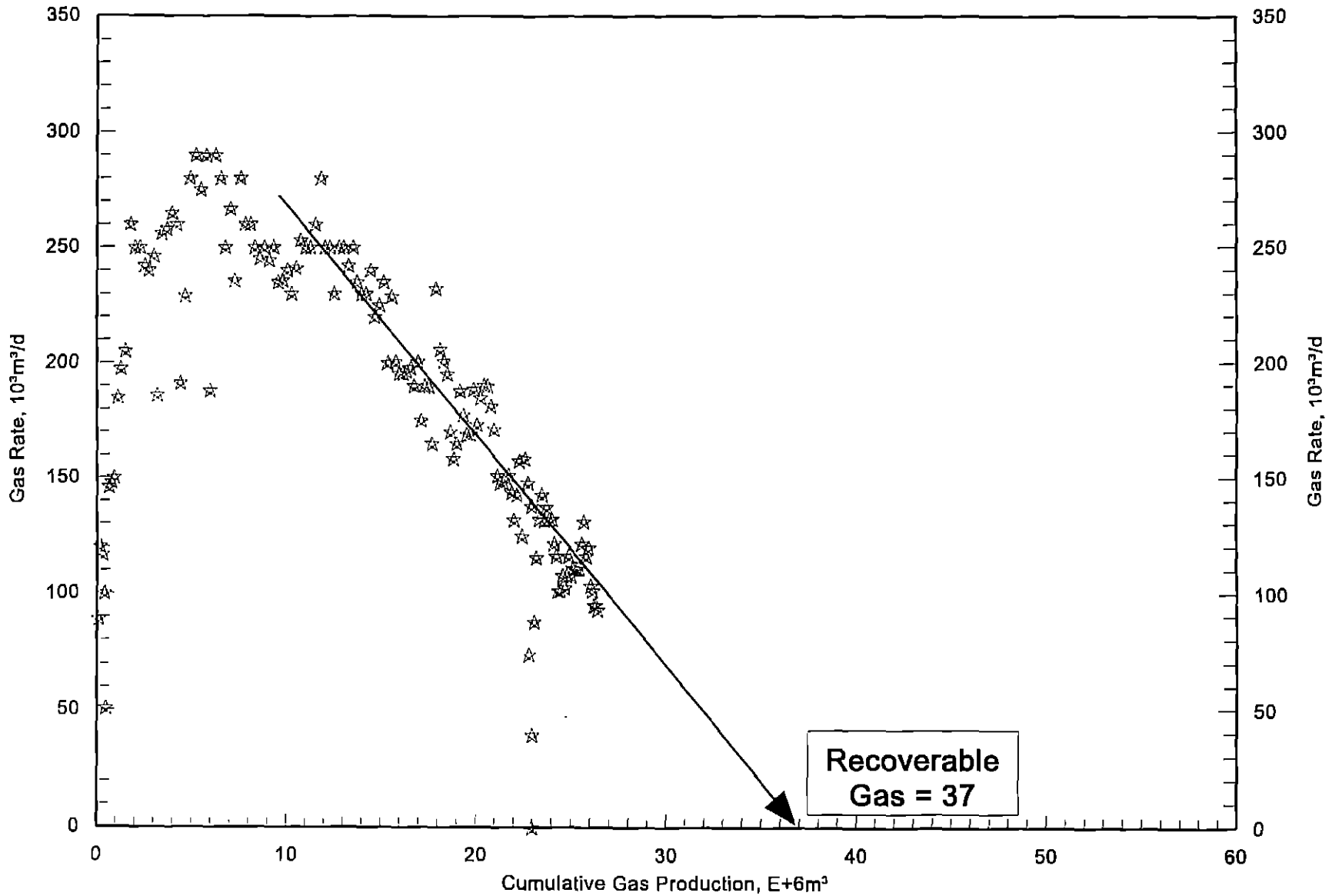

FIGURE 7: Gas Rate Decline